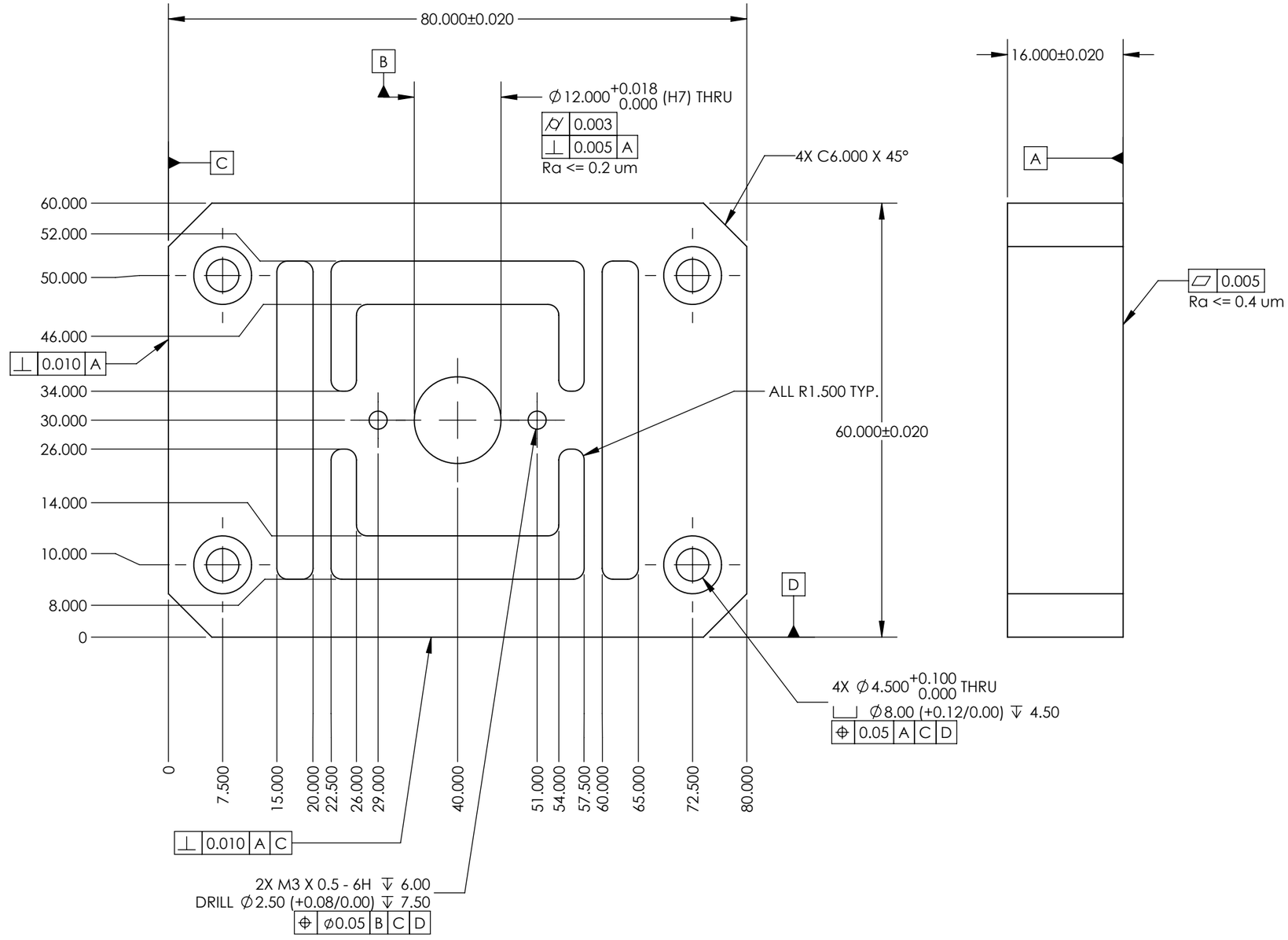
GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

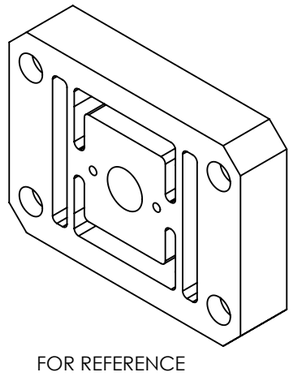
1. GENERAL & STANDARDS COMPLIANCE:
1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
1.2 3D CAD DATASET (W3-SEMI-SHUTTLE-FLEXURE-1.2.STEP) GOVERNS NOMINAL GEOMETRY FOR UNTOLERANCED SURFACES.
1.3 GENERAL TOLERANCES: LINEAR +/-0.02 mm, ANGULAR +/-0.1 DEG (HIGH-PRECISION 5-AXIS CNC / WEDM).
1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.
1.5 COORDINATE REFERENCE FRAME: FIRST OCTANT CONVENTION (ALL POSITIVE X, Y, Z FROM DATUM ORIGIN C-D-A).

2. SEMICONDUCTOR PICK-AND-PLACE & MONOLITHIC FLEXURE ASSURANCE:
2.1 MATERIAL: FORGED AEROSPACE ALUMINUM ALLOY 7075-T651 (AMS 4045, PRE-STRETCHED PLATE, YIELD STRENGTH >= 505 MPa).
2.2 100% SUBTRACTIVE 5-AXIS CNC MACHINING / WIRE-EDM ONLY. ZERO ADDITIVE MANUFACTURING (SLM/SLS) PERMITTED TO ELIMINATE AIR PARTICULATES IN CLASS 1 CLEANROOM.
2.3 MONOLITHIC FLEXURE INTEGRATION: 4-BAR PARALLEL FLEXURES ARE INTEGRALLY MACHINED WITH SOLID FRAME AND CENTRAL SHUTTLE TO ENSURE ZERO MECHANICAL BACKLASH AND ZERO FRICTION STICK-SLIP.

3. KINEMATIC RIGIDITY & GD&T CONTROLS:
3.1 PRIMARY DATUM [A] (BOTTOM MOUNTING BASE): FLATNESS <= 0.005 mm, SURFACE FINISH Ra <= 0.4 MICRONS.
3.2 SECONDARY DATUM [B] (COLLET PILOT BORE Ø12.000 +0.018/0.000 H7): CYLINDRICITY <= 0.003 mm, PERPENDICULAR TO [A] WITHIN 0.005 mm. SURFACE FINISH Ra <= 0.2 MICRONS.
3.3 2X PARALLEL FLEXURE BLADES (THICKNESS 2.50 +/-0.02 mm, SPAN 44.00 mm): ALL INTERNAL CUTOUT CORNERS CARRY SMOOTH VERTICAL FILLETS R1.50 mm TO ELIMINATE NOTCH-SENSITIVE FATIGUE NUCLEATION.
3.4 2X M3 THREADED HOLES AT (29.0, 30.0) AND (51.0, 30.0): TRUE POSITION WITHIN Ø0.05 mm RELATIVE TO [B], [C], [D]. MINIMUM SOLID ISOLATION WEB TO CENTRAL BORE >= 3.75 mm.
3.5 4X CORNER MOUNTING COUNTERBORES: CENTERED AT (7.5, 10.0) AND (72.5, 10.0) GRID, TRUE POSITION WITHIN Ø0.05 mm RELATIVE TO [A], [C], [D].

4. CLEANING & SURFACE FINISH:
4.1 DEBURR 100% UNDER 30X MAGNIFICATION. ALL SHARP EDGES BROKEN C0.20 mm MAX.
4.2 SURFACE TREATMENT: HARD ANODIZE TYPE III (MIL-A-8625, 20-25 µm THICK, PTFE SEALED FOR ULTRA-LOW PARTICULATE GENERATION).
4.3 MASKING DIRECTIVE: CENTRAL BORE Ø12.000 H7 MUST BE 100% MASKED PRIOR TO ANODIZING TO MAINTAIN SUB-MICRON SLIDING FIT.
4.4 CLEANROOM PACKAGING: ULTRASONICALLY CLEANED WITH 18 MOhm·cm DEIONIZED WATER, VACUUM SEALED FOR CLASS 1 / ISO 1 CLEANROOM COMPLIANCE.





FOR REFERENCE

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ±0.02 mm, Angular ±0.1° (High-Precision 5-Axis CNC Milling / WEDM)	FINISH: Hard Anodize Type III (MIL-A-8625, 20-25 µm, Ultra-Clean PTFE Sealed) / Critical Mating Bores Masked
DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING
REVISION: A	

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:
PRECISION DIE PICK-AND-PLACE FLEXURE ALIGNMENT SHUTTLE

MATERIAL:
Forged Aerospace Aluminum Alloy 7075-T651
(AMS 4045, Pre-Stretched Plate, Density 2.81 g/cm³)

WEIGHT: ~156.9 g

DWG NO.
W3-SEMI-SHUTTLE-FLEXURE-1.2

A3

SCALE NTS

SHEET 1 OF 1